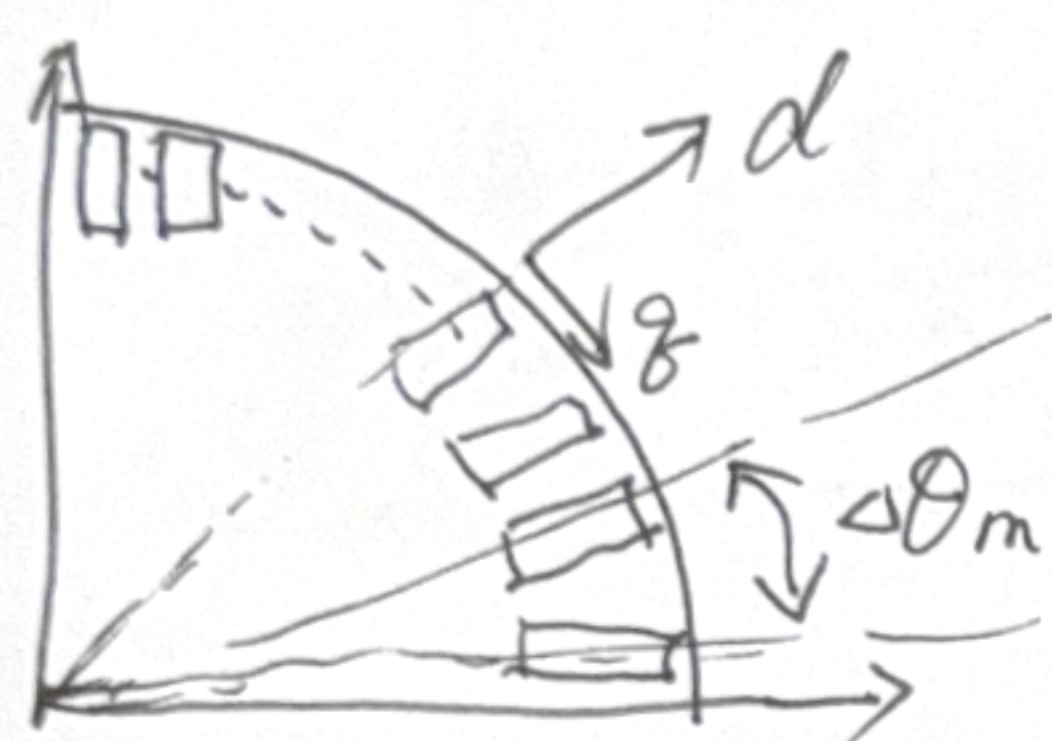


InitPos:

phase		Q (slots)	P (Pole Pairs)	m (phase)	g (slots per belt)	n (coils in sector)	InitPos	step
3f	36s	36	2	3	3	9	-30°	2
	60s	60	2	3	5	15	-30°	2
5f	40s	40	2	5	2	10	-45°	3
	60s	60	2	5	3	15	-42°	3



$$\Delta\theta_m = \frac{360^\circ}{\text{Slots}} = \frac{360^\circ}{Q}$$

$$\theta_e = \frac{360^\circ}{2Q}$$

⇒ Coil:

0,

$$\frac{360^\circ}{2Q}$$

1,

$$\frac{360^\circ}{2Q} + \Delta\theta_m = \frac{360^\circ}{Q} \left(1 + \frac{1}{2}\right)$$

2,

$$\frac{360^\circ}{2Q} + 2\Delta\theta_m = \frac{360^\circ}{Q} \left(2 + \frac{1}{2}\right)$$

⋮

k,

$$\frac{360^\circ}{2Q} + k\Delta\theta_m = \frac{360^\circ}{Q} \left(k + \frac{1}{2}\right) \quad (A)$$

e: electrical

m: mechanical

Similar to Force vectors:

$$W_A = \sum_{k \in \text{phase A}} V_k = \sum_{k \in A} \left(P_{\text{parity}_k} \cos \theta_{k,e}, P_{\text{parity}_k} \sin \theta_{k,e} \right)$$

$$Y_{A,e} = \text{atan2}(Y_A, X_A) = \text{atan2} \left(\sum_{k \in A} P_{\text{parity}_k} \sin \theta_{k,e}, \sum_{k \in A} P_{\text{parity}_k} \cos \theta_{k,e} \right)$$

$$Y_{A,m} = \frac{Y_{A,e}}{P} = 45^\circ + \text{InitPos}$$

Pole pairs d-axis angle

$$\Rightarrow \text{InitPos} = \frac{Y_{A,e}}{P} - 45^\circ \quad (B)$$

⇒ Compute $Y_{A,e}$

For 5f-60s: Q=60, P=2, m=5, g=3, step=3, n=15

According to $b = (k + \text{belt-offset}) / g = (k+1) / 3$, phase = (step × b) % m

⇒ k=0,1 ⇒ b=0, phase=0 ⇒ E(A,+)

k=2,3,4 ⇒ b=1, phase=3 ⇒ E(D,-)

k=5,6,7 ⇒ b=2, phase=1 ⇒ G(B,+)

k=8,9,10 ⇒ b=3, phase=4 ⇒ E(E,-)

k=11,12,13 ⇒ b=4, phase=2 ⇒ E(C,+)

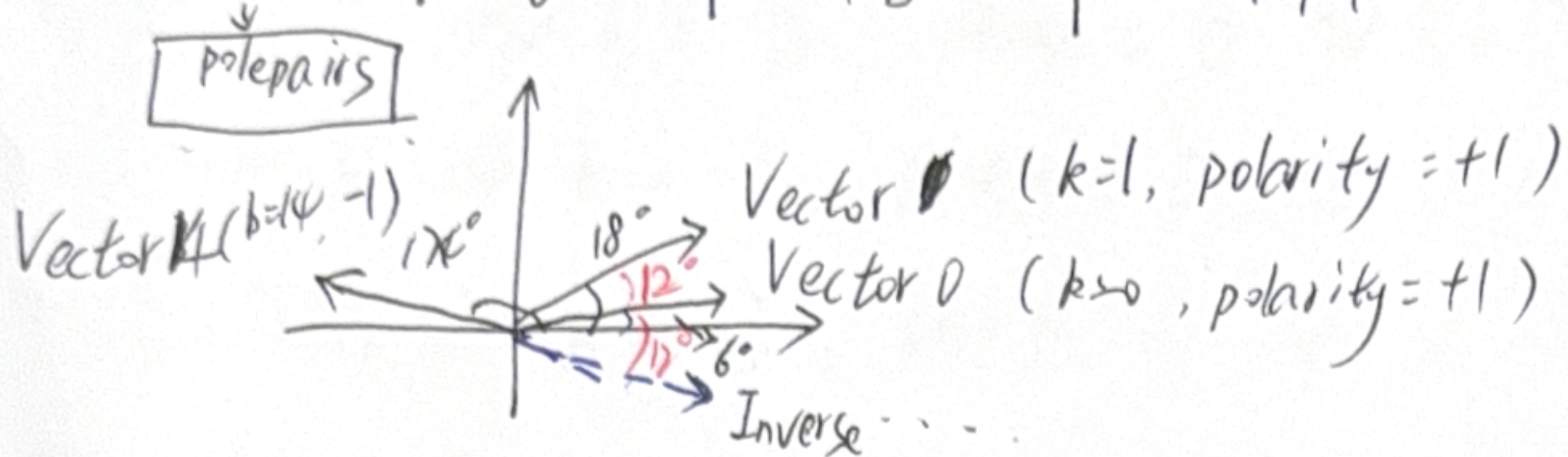
k=14 ⇒ b=5, phase=0 ⇒ E(A,-)

Ⓚ

For phase A,

R	0	1	14
polarity	+1	+1	-1
θ_m	3°	9°	87°
$\theta_e(p.\theta_m)$	6°	18°	174°

→ According to Eq. A



$$\Rightarrow Y_{A,e} = 6^\circ \quad (\text{Vector projection})$$

$$\Rightarrow \text{InitPos} = \frac{r_{A,e}}{p} - 45^\circ = \frac{6^\circ}{2} - 45^\circ = -42^\circ$$

Similarly,

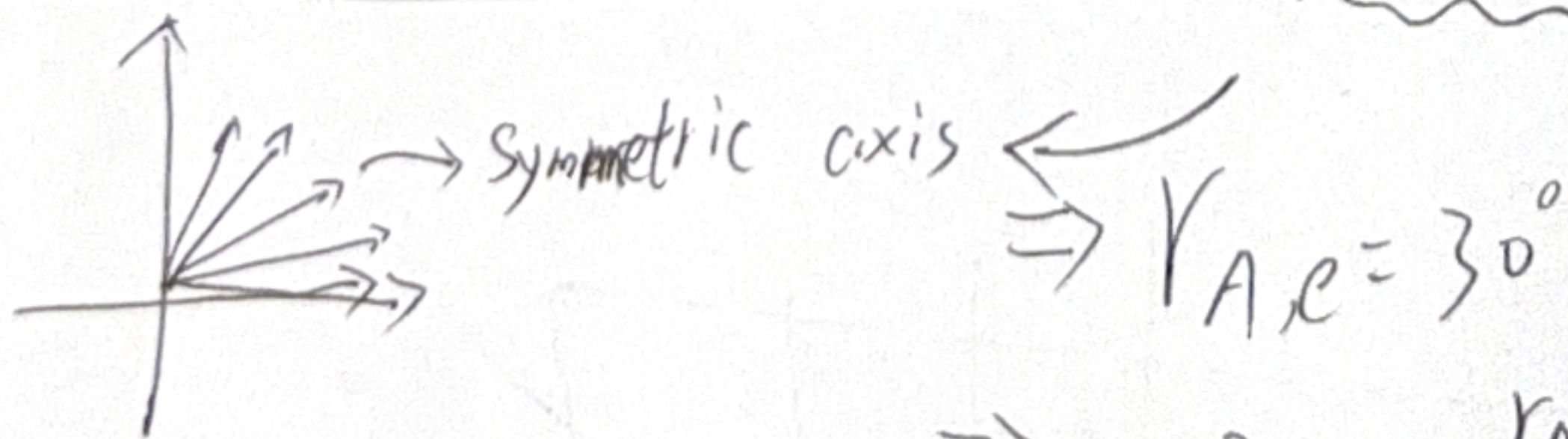
For 3f-6os; $Q=60$, $P=2$, $m=3$, $g=5$, $\text{step}=2$, $n=15$

According to $b = (k + \text{belt_offset}) \parallel g = (k + 0) \parallel 5$, $\text{phase} = (\text{step} \times b) \% m$
 $\Rightarrow k = 0, 1, 2, 3, 4 \Rightarrow b = 0, \text{phase} = 0 \Rightarrow \in(A, +) = (2b) \% 3$

$$k=5, 6, 7, 8, 9 \Rightarrow b=1, \text{ phase}=2 \Rightarrow \mathcal{H}(C, -)$$
$$k=10, 11, 12, 13, 14 \Rightarrow b=2, \text{ phase}=1 \Rightarrow \in (\mathbb{R}, +)$$

$\Rightarrow$ For phase A

k	0	1	2	3	4	
polarity	+1	+1	+1	+1	+1	
θ_m	3°	9°	15°	21°	27°	
θ_e	6°	18°	30°	42°	54°	



$$\Rightarrow \text{InitPos} = \frac{r_{A,e}}{p} - 45^\circ = \frac{30^\circ}{2} - 45^\circ = -30^\circ$$

Similarly,

For 3f-36s, $Q=36$, $P=2$, $m=3$, $g=3$, $step=2$, $\eta=9$

$$b = (k + belt_offset) // g = (k + 0) // 3$$

$$phase = (step \times b) \% m = (2b) \% 3$$

$$\Rightarrow k=0, 1, 2 \Rightarrow b=0, phase=0 \Rightarrow E(A, +)$$

$$k=3, 4, 5 \Rightarrow b=1, phase=2 \Rightarrow E(C, -)$$

$$k=6, 7, 8 \Rightarrow b=2, phase=1 \Rightarrow E(B, +)$$

$\Rightarrow$ For phase A

k	0	1	2
polarity	+1	+1	+1
θ_m	5°	15°	25°
θ_e	10°	30°	50°

$\nwarrow$ symmetric axis

$$\Rightarrow \gamma_{A,e} = 30^\circ$$

$$\Rightarrow \text{InitPos} = \frac{\gamma_{A,e}}{P} - 45^\circ = \frac{30^\circ}{2} - 45^\circ = -30^\circ$$

3f-36s

For 5f-40s, $Q=40$, $P=2$, $m=5$, $g=8$, $step=3$, $\eta=10$

$$b = (k + belt_offset) // g = (k + 1) // 8$$

$$phase = (step \times b) \% m = (3b) \% 5 = (3b) \% 5$$

$$\Rightarrow k=0 \Rightarrow b=0, phase=0 \Rightarrow E(A, +)$$

$$k=1, 2 \Rightarrow b=1, phase=3 \Rightarrow E(D, -)$$

$$k=3, 4 \Rightarrow b=2, phase=1 \Rightarrow E(B, +)$$

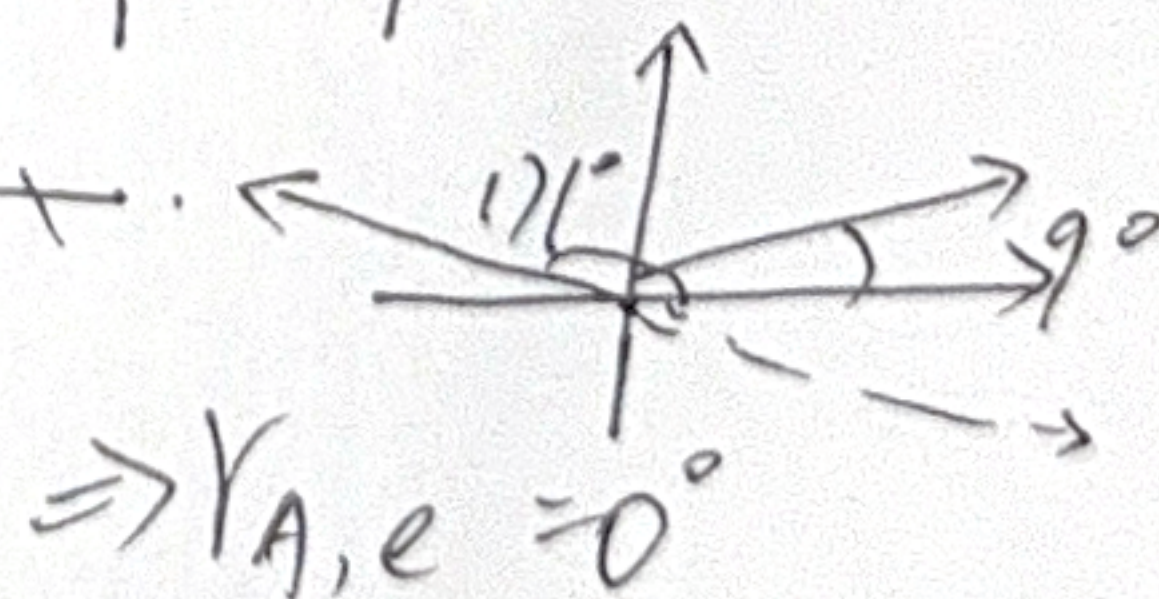
$$k=5, 6 \Rightarrow b=3, phase=4 \Rightarrow E(E, -)$$

$$k=7, 8 \Rightarrow b=4, phase=2 \Rightarrow E(C, +)$$

$$k=9 \Rightarrow b=5, phase=0 \Rightarrow E(A, -)$$

$$\Rightarrow$$

k	0	9
polarity	+1	-1
θ_m	4.5°	85.5°
θ_e	9°	171°



$$\Rightarrow \gamma_{A,e} = 0^\circ$$

$$\Rightarrow \text{InitPos} = \frac{\gamma_{A,e}}{P} - 45^\circ = \frac{0^\circ}{2} - 45^\circ = -45^\circ$$

5f-40s